CHAPTER 14 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

Cybersecurity of High-Altitude Platform Systems: Anti-Jamming and Resilient Communications

High-Altitude Platform Systems

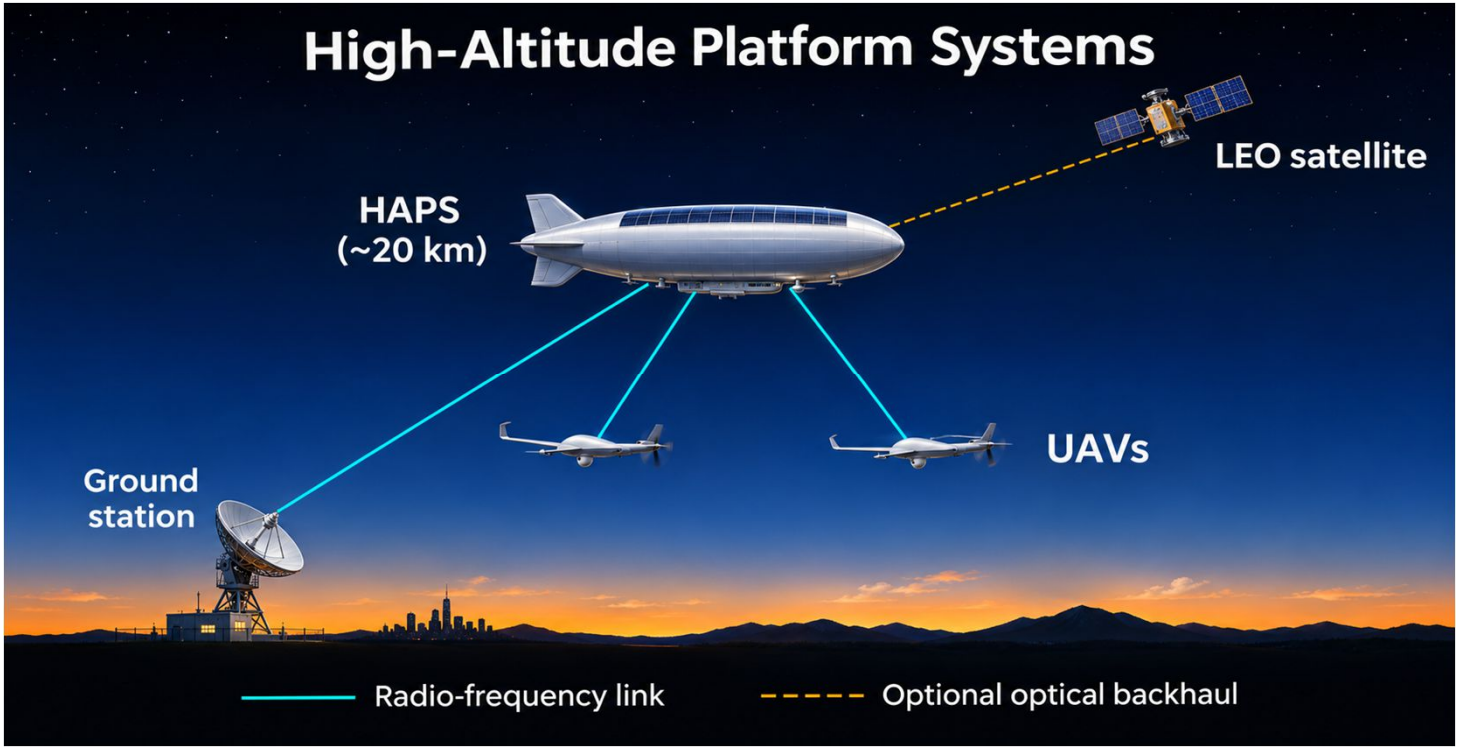
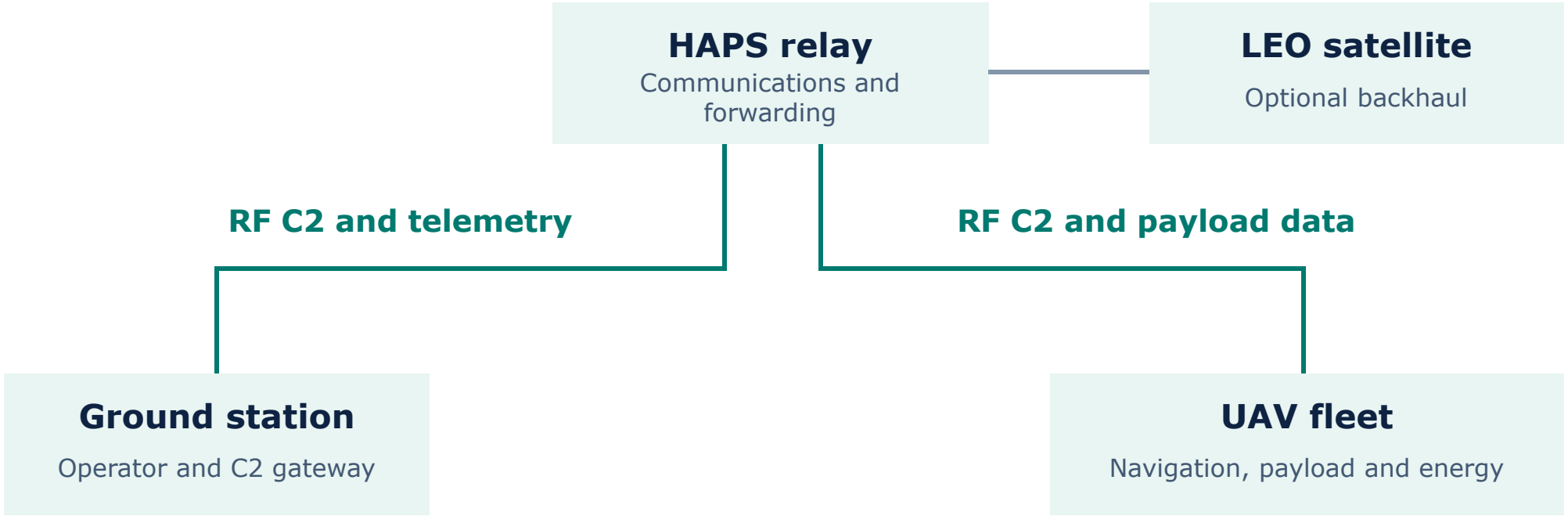


Figure S14.1. High-altitude platform system (HAPS) supporting UAV communications. A stratospheric airship operating at approximately 20 km relays traffic between lower-altitude unmanned aerial vehicles and a terrestrial ground station over radio-frequency links (solid cyan). An optional free-space optical link to a low Earth orbit (LEO) satellite provides a separate backhaul path (dashed amber). The illustration is schematic and not to scale.

Threat Models for HAPS and Near-Space Platforms

HAPS–UAV threat model schematic



Stale commands can affect the gateway.

Threats include interference and navigation faults.

HAPS = high-altitude platform system. LEO = low Earth orbit. C2 = command and control.

Threat Models for HAPS and Near-Space Platforms

System scope and threat scenarios

$$\mathcal{S}_H = \{\mathcal{C}, \mathcal{N}, \mathcal{P}, \mathcal{E}, \mathcal{G}, \mathcal{U}\}, \tag{14.1}$$

$$\mathcal{A} = \{A_0, A_1, A_2, A_3, A_4, A_5\}, \tag{14.2}$$

Exercise notation. C communications, N navigation, P payload, E energy, G ground segment, U UAV fleet.

Case	Simulated condition	Primary elements affected
A0	Nominal operation	Baseline across all elements
A1	RF interference	C, U
A2	Stale commands at the gateway	C, G, U
A3	Inconsistent navigation solution	N, U
A4	Suspect payload data	P, U
A5	Reduced energy reserve	E, U

Threat Models for HAPS and Near-Space Platforms

Link quality and C2 availability

Signal-to-interference-plus-noise ratio (SINR)

$$\mathrm{SINR}_H = \frac{P_s G_s}{N_0 B + P_j G_j + \sum_{k=1}^{K} P_k G_k}, \tag{14.3}$$

Use received powers in pW. Convert the linear ratio to dB using $10 \log_{10}(\mathrm{SINR})$.

Usable command-and-control time

$$A_{C2} = 1 - \frac{T_{\mathrm{loss}} + T_{\mathrm{stale}} + T_{\mathrm{unsafe}}}{T_{\mathrm{mission}}}, \tag{14.4}$$

The log classifies unavailable time as loss, stale commands, or unsafe operation. These intervals are mutually exclusive.

Threat Models for HAPS and Near-Space Platforms

Mission trust score

$$\tau_H = \alpha Q_{C2} + \beta Q_N + \gamma Q_P + \delta Q_E + \eta Q_G, \quad \alpha + \beta + \gamma + \delta + \eta = 1, \quad (14.6)$$

All quality inputs range from 0 to 1. For this exercise, set C2 quality equal to calculated C2 availability.

Quality input	Weight	Exercise interpretation
C2	0.35	Fraction of mission time with usable C2
Navigation	0.20	Consistency of the navigation solution
Payload	0.15	Quality and integrity of collected data
Energy	0.15	Adequacy of power reserves
Ground segment	0.15	Confidence in gateway and operator-system integrity

Threat Models for HAPS and Near-Space Platforms

Simulated mission data

Mission time 600 s. Desired received power 100 pW. Noise 5 pW. Other interference 5 pW.

Case	Added interference (pW)	Loss (s)	Stale (s)	Unsafe (s)
A0	0	6	0	0
A1	40	90	18	12
A2	0	0	120	0
A3	0	0	0	60

Case	Navigation	Payload	Energy	Ground segment
A0	0.95	0.96	0.95	0.96
A1	0.92	0.90	0.92	0.95
A2	0.95	0.96	0.95	0.40
A3	0.45	0.94	0.90	0.92

Threat Models for HAPS and Near-Space Platforms

Student tasks and decision criteria

1. Calculate	For A0–A3, calculate SINR in dB, C2 availability, and mission trust. Use the unrounded values in the decision checks.
2. Compare the rules	Apply the SINR rule first, then the joint rule. List every failed condition and explain why a good RF link can coexist with unusable C2.
3. Test recovery	In A2, reduce stale time to 30 s and raise ground quality to 0.85. Recalculate and reassess. Propose one monitor each for A4 and A5.

SINR-only rule. Continue when SINR ≥ 6 dB. Otherwise pause ISR.

Joint rule. Continue only when SINR ≥ 6 dB, C2 availability ≥ 0.90, trust ≥ 0.82, navigation ≥ 0.60, and ground quality ≥ 0.70. Otherwise pause ISR for review.

Submit a decision table and a short explanation of what the A2 case reveals.

Threat Models for HAPS and Near-Space Platforms

Scores and mission decisions

Case	SINR (dB)	Availability	Trust	SINR only	Joint rule
A0	10.00	0.99	0.9670	Continue	Continue
A1	3.01	0.80	0.8795	Pause	Pause
A2	10.00	0.80	0.8165	Continue	Pause
A3	10.00	0.90	0.8190	Continue	Pause

Failed joint conditions

A1 SINR, Availability

A2 Availability, Trust, Ground trust

A3 Trust, Navigation

Threat Models for HAPS and Near-Space Platforms

A2 diagnosis and recovery

A2 has 10.00 dB SINR, but stale commands reduce usable C2 time to 0.80. The joint rule pauses ISR.

Recovered availability = 1 − 30 / 600 = 0.95

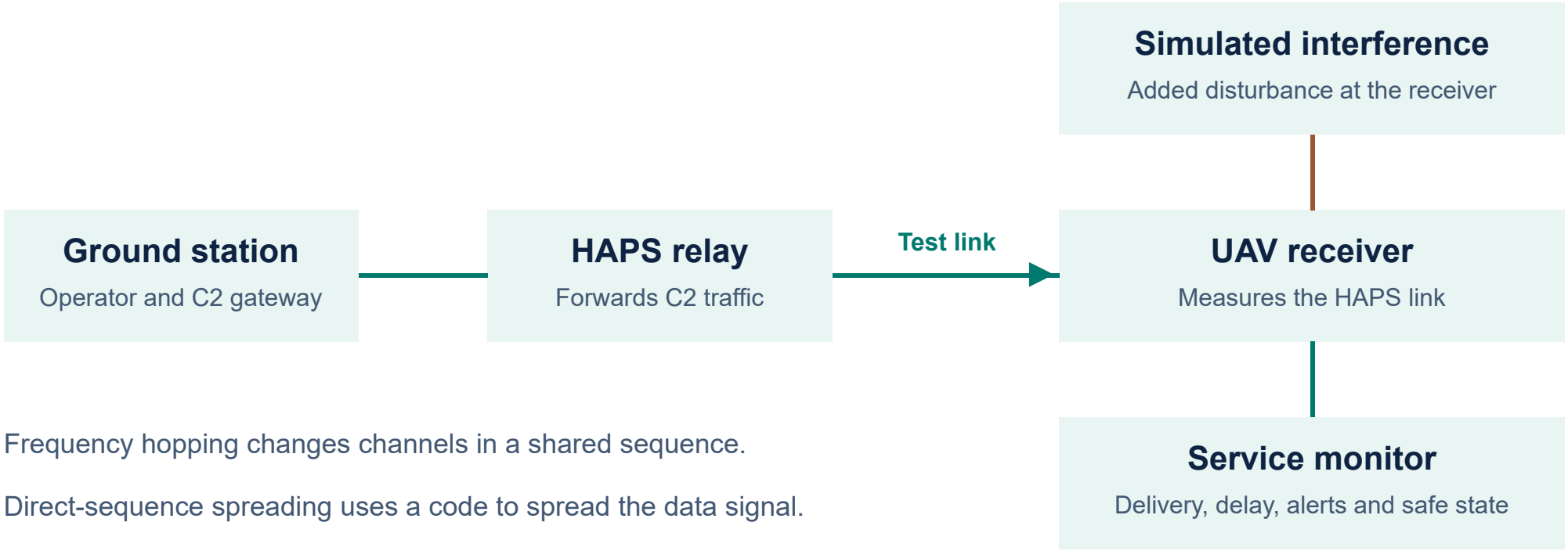
Recovered trust = 0.35 × 0.95 + 0.20 × 0.95 + 0.15 × 0.96 + 0.15 × 0.95 + 0.15 × 0.85 = 0.9365.

The link remains at 10.00 dB. All joint thresholds now pass, so the simulated ISR task can resume.

A4 Verify payload data signatures.
A5 Track remaining energy against a reserve threshold.

Anti-Jamming and Spectrum-Resilient Communication Techniques

HAPS–UAV link and monitoring schematic



Frequency hopping changes channels in a shared sequence.

Direct-sequence spreading uses a code to spread the data signal.

HAPS = high-altitude platform system. C2 = command and control.

Anti-Jamming and Spectrum-Resilient Communication Techniques

Link margin and spreading gain

$$\mathrm{SINR}_{h,u}(t) = \frac{P_h G_{h,u}(t)}{N_0 B + \sum_{j=1}^{J} P_j G_{j,u}(t) + \sum_{k=1}^{K} P_k G_{k,u}(t)}, \qquad (14.8)$$

$$\mathrm{SINR}_{h,u}(t) > \Gamma_{\min}, \qquad (14.9)$$

Convert SINR to dB with $10 \log_{10}(\mathrm{SINR})$. The link passes when SINR exceeds 6 dB.

$$G_p \approx \frac{B_{\mathrm{spread}}}{R_b}, \qquad G_{p,\mathrm{dB}} = 10 \log_{10}(G_p), \qquad (14.10)$$

For direct-sequence spread spectrum, use a spread bandwidth of 4 MHz and a bit rate of 0.5 Mb/s. Compute the nominal processing gain as a ratio and in dB.

The mode data already reflect receiver effects. Calculate nominal processing gain separately.

Anti-Jamming and Spectrum-Resilient Communication Techniques

Spectrum-resilience score

$$\rho_{\mathrm{spec}} = \omega_1 \overline{\mathrm{PDR}} + \omega_2 \overline{A}_{\mathrm{svc}} + \omega_3 (1 - \overline{D}_{\mathrm{norm}}) + \omega_4 (1 - \overline{F}_{\mathrm{alarm}}) + \omega_5 \overline{S}_{\mathrm{safe}}, \tag{14.17}$$

All inputs are window summaries on a 0–1 scale. The five weights sum to 1.

Metric	Weight	Meaning in the exercise
Packet delivery ratio (PDR)	0.25	Fraction of expected packets delivered
Service availability	0.25	Fraction of time C2 service meets its requirements
Normalized delay	0.15	Delay penalty, with higher values indicating worse delay
False-alarm rate	0.10	False alerts as a fraction of benign evaluation intervals
Safe-state score	0.25	Fraction of the window in an acceptable simulated state

Anti-Jamming and Spectrum-Resilient Communication Techniques

Simulated link and service data

Desired signal 100 pW, noise 5 pW, other interference 5 pW. The table gives equivalent added interference for each mode at a common receiver reference.

Mode	Added J (pW)	PDR	Avail.	Delay	False alarms	Safe state
A. Fixed channel, baseline	0	0.99	0.99	0.10	0.02	1.00
B. Fixed channel, interference	40	0.68	0.72	0.55	0.03	0.80
C. Frequency hopping	10	0.94	0.96	0.22	0.04	0.98
D. Direct-sequence spread spectrum	8	0.96	0.90	0.40	0.06	0.99

All rows represent equal-duration simulation windows. Use the table values directly in the link and resilience equations.

Anti-Jamming and Spectrum-Resilient Communication Techniques

Student tasks and decision rules

1. Compute link metrics	Calculate SINR in dB for A–D. For D, also calculate the nominal spreading gain using Equation 14.10.
2. Score service	Compute the spectrum-resilience score and compare the radio-only and joint decisions for each mode.
3. Test recovery	For D, change availability to 0.96 and normalized delay to 0.25. Keep the other inputs fixed and reassess.

Radio-only rule. Pass when SINR > 6 dB.

Joint rule. Continue the simulated C2 service only if the radio rule passes, resilience ≥ 0.90, availability ≥ 0.95, and safe-state score ≥ 0.95. Otherwise pause for review.

Submit a comparison table and explain any case that passes the radio rule but fails the joint rule.

Anti-Jamming and Spectrum-Resilient Communication Techniques

Link results and service decisions

Mode	SINR (dB)	Resilience	Radio rule	Joint rule
A	10.00	0.9780	Pass	Continue
B	3.01	0.7145	Fail	Pause
C	6.99	0.9330	Pass	Continue
D	7.45	0.8965	Pass	Pause

B fails SINR, resilience, availability and safe-state requirements.

D passes the radio rule but fails resilience and availability.

For C, SINR = $10 \log_{10}[100 / (5 + 10 + 5)] = 6.99$ dB.

Anti-Jamming and Spectrum-Resilient Communication Techniques

Spreading gain and recovery

Nominal processing gain for D

4 MHz / 0.5 Mb/s = 8. In dB, 10 $\log_{10}(8) \approx 9.03$ dB.

The equivalent interference input already includes receiver effects. SINR remains 7.45 dB.

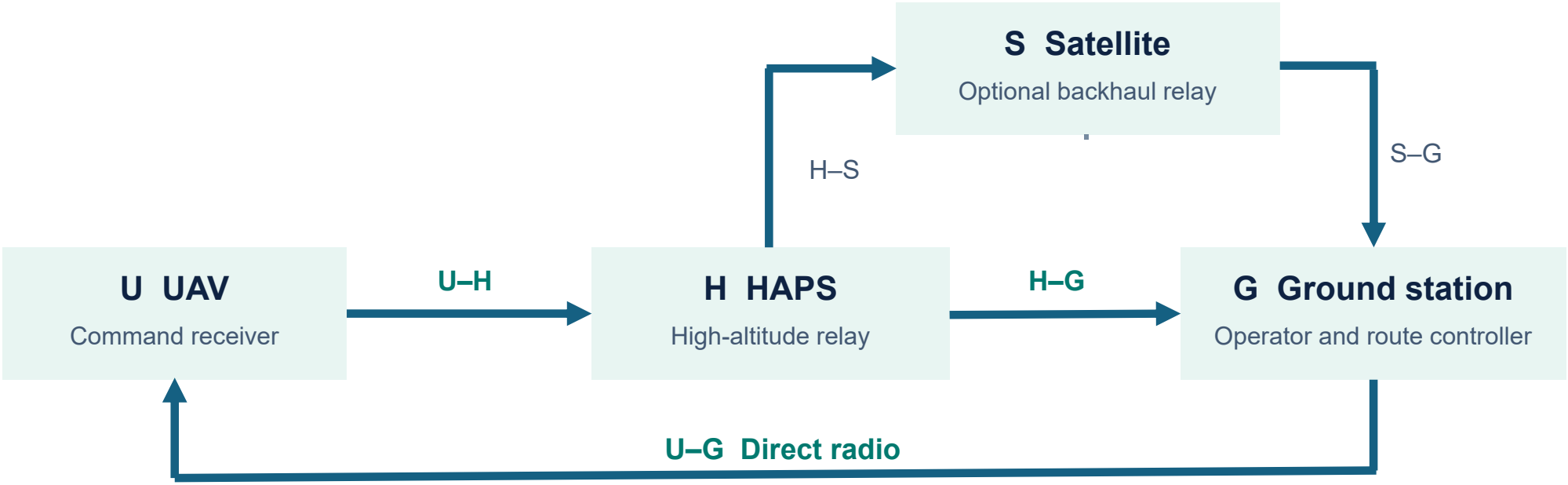
Updated service metrics for D

$0.25 \times 0.96 + 0.25 \times 0.96 + 0.15 \times 0.75 + 0.10 \times 0.94 + 0.25 \times 0.99 = 0.9340$

Availability rises to 0.96 and the safe-state score stays at 0.99. With unchanged SINR and a resilience score of 0.9340, the simulation passes the joint rule.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Secure network schematic



HAPS = high-altitude platform system. Endpoints use authenticated encryption, and the UAV checks command age and sequence.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Route risk

$$R(\pi) = \sum_{e_{ij} \in \pi} w_{ij} \left(p_{ij}^{DoS} + p_{ij}^{MITM} + p_{ij}^{jam} + p_{ij}^{spoof} + p_{ij}^{comp} \right), \qquad (14.25)$$

DoS = denial of service. MITM = man-in-the-middle. Comp. = compromise. Treat the sum as a relative score for comparing routes.

Link	Weight	DoS	MITM	Jamming	Spoofing	Comp.
U–G	1.0	0.08	0.04	0.10	0.04	0.02
U–H	1.0	0.02	0.01	0.04	0.01	0.01
H–G	1.2	0.03	0.02	0.03	0.01	0.01
H–S	1.0	0.02	0.01	0.01	0.01	0.01
S–G	1.0	0.02	0.01	0.02	0.01	0.01

Calculate the score for U–G, U–H–G, and U–H–S–G. Select the route with the lowest score.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Freshness and authentication

$$|t_{\mathrm{rx}}(m_k) - t_{\mathrm{tx}}(m_k)| \le \Delta t_{\max}, \quad seq(m_k) > seq(m_{k-1}), \quad \mathrm{Auth}(m_k) = 1, \tag{14.26}$$

Assume synchronized clocks, a 200 ms age limit, and last accepted sequence 40. Process the rows in arrival order for one authenticated command stream.

Arrival	Age (ms)	Sequence	Authentication
M1	80	41	Pass
M2	310	44	Pass
M3	90	41	Pass
M4	60	43	Fail
M5	200	42	Pass

Mark each message accepted or rejected and explain why. Update the sequence state only on acceptance.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Relay trust and safe availability

Relay eligibility

$$\tau_i(t+1) = \lambda \tau_i(t) + (1-\lambda)\left[1 - A_i(t)\right], \quad 0 \le \lambda \le 1, \tag{14.27}$$

H starts at trust 0.90. Use λ = 0.75 and anomaly score 0.60. Recalculate trust and exclude routes through any relay below 0.80. Assume S remains eligible and the direct link is usable.

Mission availability

$$A_{\text{safe}} = 1 - \frac{T_{\text{loss}} + T_{\text{stale}} + T_{\text{unauth}} + T_{\text{unsafe}}}{T_{\text{mission}}}, \tag{14.28}$$

For a 600 s mission, use 24 s lost, 18 s stale, 6 s unauthorized, and 12 s unsafe. The intervals do not overlap. Calculate safe availability and compare it with the 0.95 target.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Traffic priority under congestion

$$P_f = \omega_1 S_f + \omega_2 U_f + \omega_3 C_f - \omega_4 B_f, \qquad (14.29)$$

S = safety significance, U = urgency, C = mission criticality, B = bandwidth cost. All inputs use a 0–1 scale. Weights are 0.40, 0.30, 0.20, and 0.10. C2 = command and control.

Flow	S	U	C	B
Emergency alert	1.00	1.00	0.95	0.10
C2 command	0.95	0.90	1.00	0.20
Payload video	0.20	0.40	0.30	0.90

Rank the flows by descending score. If the next scheduling window can serve only two flows, identify which two to send.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Route selection and relay trust

Route	Risk calculation	Score	After trust update
Direct	0.28	0.28	Eligible
HAPS relay	0.09 + 0.12	0.21	Ineligible
Satellite backhaul	0.09 + 0.06 + 0.07	0.22	Ineligible

Initially choose U–H–G. Its risk score of 0.21 is the lowest.

Updated H trust = $0.75 \times 0.90 + 0.25 \times (1 - 0.60) = 0.775$.

Both relay routes pass through H and become ineligible. Select the direct U–G link with risk 0.28.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Message acceptance and sequence state

Arrival	Decision	Reason	Last accepted sequence
M1	Accept	All checks pass	41
M2	Reject	Age exceeds 200 ms	41
M3	Reject	Sequence already accepted	41
M4	Reject	Authentication fails	41
M5	Accept	All checks pass	42

M5 passes at exactly 200 ms. Its sequence 42 exceeds the last accepted sequence, 41.

Rejected messages M2 and M4 leave the receiver state unchanged even though their sequence numbers are higher.

Secure Networking Between UAVs, HAPS, Satellites, and Ground Stations

Availability and scheduling decisions

Safe availability = 1 − (24 + 18 + 6 + 12) / 600 = 0.90.

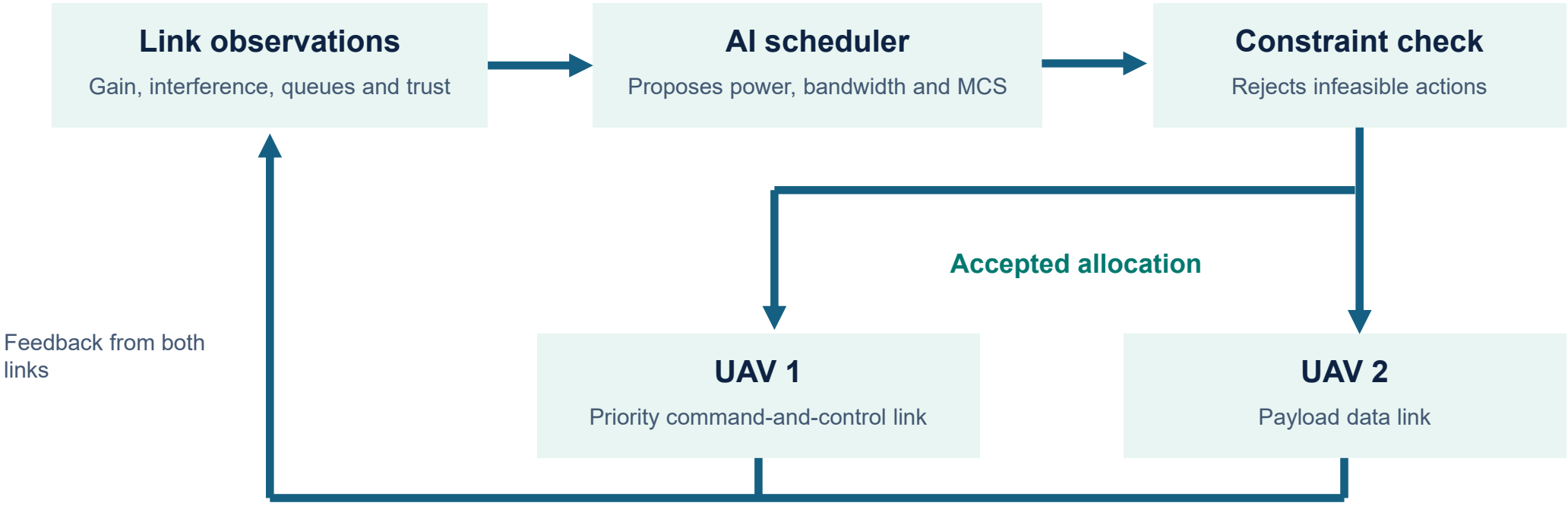
The mission misses the 0.95 safe-availability target by 5 percentage points.

Rank	Flow	Priority	Next window
1	Emergency alert	0.88	Send
2	C2 command	0.83	Send
3	Payload video	0.17	Defer

Emergency alert = 0.40 × 1.00 + 0.30 × 1.00 + 0.20 × 0.95 − 0.10 × 0.10 = 0.88.

AI-Based Resource Allocation and Link Adaptation

AI scheduling and link adaptation schematic



MCS = modulation and coding scheme. This exercise fixes the MCS on each link and compares three proposed allocations.

AI-Based Resource Allocation and Link Adaptation

Link quality and rate model

$$\mathrm{SINR}_{u,t} = \frac{P_{u,t} G_{u,t}}{N_0 B_{u,t} + J_{u,t} + I_{u,t}}, \qquad (14.32)$$

$$R_{u,t} = B_{u,t} \log_2 \left(1 + \mathrm{SINR}_{u,t}\right). \qquad (14.33)$$

Use linear SINR, power in mW, and bandwidth in MHz. Noise density N_0 = 0.002 mW/MHz. The calculated rate is an idealized value in Mb/s.

Link	Gain G	Jamming J (mW)	Other interference I (mW)	Min. SINR Γ
U1	0.010	0.008	0.002	3.00
U2	0.010	0.002	0.002	2.00

Calculate SINR and rate for both links under every candidate allocation.

AI-Based Resource Allocation and Link Adaptation

Allocation objective and constraints

$$\max_{\mathbf{a}_t} \sum_{u=1}^{U} w_u \left[\alpha R_{u,t} - \beta D_{u,t} - \gamma E_{u,t} - \delta \rho_{u,t} + \lambda T_{u,t}\right], \tag{14.34}$$

subject to

$$\sum_{u=1}^{U} P_{u,t} \le P_{\max}, \quad \sum_{u=1}^{U} B_{u,t} \le B_{\max}, \quad D_{u,t} \le D_u^{\max}, \quad \mathrm{SINR}_{u,t} \ge \Gamma_{\mu_{u,t}}, \tag{14.35}$$

Total power ≤ 8 mW and total bandwidth ≤ 2 MHz. Delay limits are 60 ms for U1 and 100 ms for U2. Use the minimum SINR on slide 3.

User weights w are 2 for U1 and 1 for U2. Use α = 1, β = 10, γ = 0.40, δ = 1, and λ = 0.50. Convert delay to seconds for the objective.

Record all violated constraints. Select the feasible candidate with the largest objective value.

AI-Based Resource Allocation and Link Adaptation

Candidate allocations and observations

Each pair is ordered (U1, U2). Slot duration is 0.10 s, so energy per link is power × 0.10 s in mJ. Trust is (0.80, 0.85) for all candidates.

Action	Power (mW)	Bandwidth (MHz)	Delay (ms)	Attack risk ρ
A. Balanced	4, 4	1.0, 1.0	58, 60	0.40, 0.15
B. Favor U1	6, 2	1.2, 0.8	38, 85	0.25, 0.20
C. Over budget	7, 3	1.5, 1.0	28, 110	0.20, 0.20

Treat delay and attack risk as simulated predictions for each action. U1 carries priority command and control. U2 carries payload data.

AI-Based Resource Allocation and Link Adaptation

Reinforcement learning reward

$$r_t = \sum_{u=1}^{U} \left(\omega_1 \frac{R_{u,t}}{R_u^{\max}} - \omega_2 \frac{D_{u,t}}{D_u^{\max}} - \omega_3 \frac{E_{u,t}}{E_u^{\max}} - \omega_4 \rho_{u,t} + \omega_5 \sigma_{u,t} \right) - \omega_6 \mathbb{I}_{\text{unsafe}}, \tag{14.36}$$

Rate references are (5, 4) Mb/s. Delay references are (60, 100) ms. Energy references are (0.80, 0.80) mJ. The normalized terms use consistent units.

Use ω_1 through ω_6 = (0.35, 0.20, 0.10, 0.15, 0.20, 1.00).

Action	Safety indicators (σ_1, σ_2)	Unsafe-event indicator
A	1, 1	0
B	1, 1	0
C	1, 0	1

Compute rewards offline. Among feasible actions, compare the reward ranking with the objective ranking. Explain why the feasibility check still applies.

AI-Based Resource Allocation and Link Adaptation

Trust update and secrecy rate

$$T_{u,t+1} = \kappa T_{u,t} + (1-\kappa)\left[\theta_1 \mathrm{PDR}_{u,t} + \theta_2 S_{u,t} + \theta_3 A_{u,t} - \theta_4 \rho_{u,t}\right], \qquad (14.39)$$

For U1 under B, use prior trust 0.80, κ = 0.75, and θ_1–θ_4 = (0.40, 0.35, 0.25, 0.20). Packet delivery ratio = 0.94, safety evidence S = 0.90, availability A = 0.96, and attack risk ρ = 0.25. Find next-slot trust.

$$R_{s,u,t} = \left[B_{u,t}\log_2\left(1+\mathrm{SINR}_{u,t}^{\mathrm{legit}}\right) - B_{u,t}\log_2\left(1+\mathrm{SINR}_{u,t}^{\mathrm{adv}}\right)\right]^+, \qquad (14.40)$$

Use U1's bandwidth and legitimate SINR under B. Evaluate secrecy rate for adversary SINR of 2.50 and 5.00, both linear. Which case meets a minimum secrecy rate of 0.80 Mb/s?

AI-Based Resource Allocation and Link Adaptation

Feasibility, objective, and reward

Action	SINR (U1, U2)	Rate (Mb/s)	Objective	Reward	Constraint violations
A	3.33, 6.67	2.12, 2.94	5.205	0.309	None
B	4.84, 3.57	3.05, 1.75	6.219	0.303	None
C	5.38, 5.00	4.01, 2.58	8.894	-0.791	Power, bandwidth, U2 delay

Select B. It has the largest objective among feasible candidates A and B.

For U1 under B, SINR = 0.060 / (0.0024 + 0.008 + 0.002) ≈ 4.84.

C's raw objective is higher, but its power, bandwidth, and U2 delay exceed their limits.

A has the higher reward. The reward normalizes both links and omits the objective's extra weight for U1.

AI-Based Resource Allocation and Link Adaptation

Worked trust and secrecy calculations

U1 trust for the next slot

$$0.75 \times 0.80 + 0.25 \times (0.40 \times 0.94 + 0.35 \times 0.90 + 0.25 \times 0.96 - 0.20 \times 0.25) = 0.82025$$

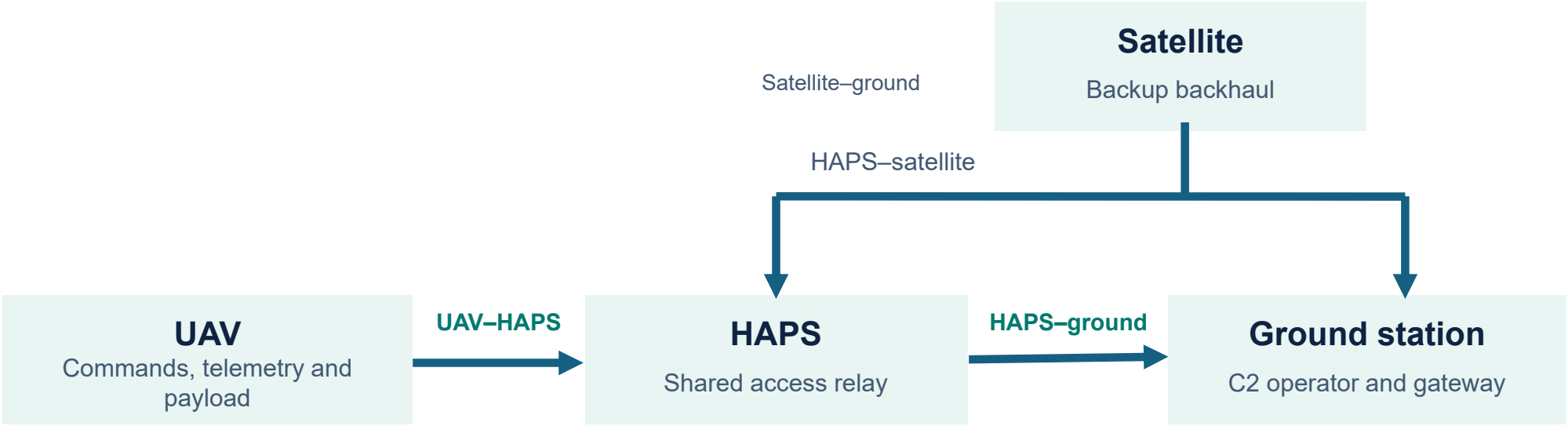
U1 secrecy rate under B

Adversary SINR	Secrecy rate (Mb/s)	Meets 0.80 Mb/s?
2.50	0.886	Yes
5.00	0.000	No

At adversary SINR 5.00, the adversary's modeled rate exceeds the legitimate rate. The positive-part operator sets secrecy rate to zero.

HAPS as Resilient Communication Relays for UAV Operations

HAPS relay and backup schematic



Both backhaul routes share the UAV–HAPS access link.

HAPS = high-altitude platform system. C2 = command and control.

HAPS as Resilient Communication Relays for UAV Operations

System model and access-link SINR

$$\mathcal{R}_{HU} = \{L_{u,h}, L_{h,g}, L_{h,s}, \mathcal{C}_u, \mathcal{T}_u, \mathcal{P}_u\}, \qquad (14.41)$$

The three L terms identify the links labeled on slide 2. In this exercise, C_u, T_u, and P_u denote commands, telemetry, and payload traffic. The satellite-to-ground leg completes the backup route.

$$\mathrm{SINR}_{u,h} = \frac{P_u G_{u,h}}{N_0 B + P_j G_{j,h} + \sum_{k=1}^{K} P_k G_{k,h}}, \qquad (14.42)$$

For every case, desired received power is 100 pW, noise is 5 pW, and other interference is 5 pW. Use the added interference in the table on slide 5.

Convert the linear SINR ratio to dB with $10 \log_{10}(\mathrm{SINR})$.

HAPS as Resilient Communication Relays for UAV Operations

C2 availability and mission trust

$$A_{HU} = 1 - \frac{T_{loss} + T_{stale} + T_{unsafe}}{T_{mission}} \qquad (14.43)$$

Each case covers 600 s. The logged loss, stale-command, and unsafe intervals are mutually exclusive.

$$\tau_{HU} = \alpha Q_{\text{link}} + \beta Q_{\text{fresh}} + \gamma Q_{\text{auth}} + \delta Q_{\text{route}} + \eta Q_{\text{payload}}, \quad \alpha + \beta + \gamma + \delta + \eta = 1, \qquad (14.45)$$

The trust inputs represent access-link quality, message freshness, authentication, route health, and payload integrity. Scores range from 0 to 1 and are supplied separately from SINR.

Use weights (0.25, 0.25, 0.20, 0.20, 0.10) in equation order.

HAPS as Resilient Communication Relays for UAV Operations

Simulated relay cases

C and D represent the same ground-backhaul outage before and after switching to satellite backup.

Case	Condition	Added J (pW)	Lost (s)	Stale (s)	Unsafe (s)
A	Nominal ground backhaul	0	6	0	0
B	UAV access interference	40	90	18	12
C	Ground backhaul outage	0	102	24	6
D	Satellite backup active	0	18	12	0

Case	Link	Freshness	Authentication	Route	Payload
A	0.95	0.98	1.00	0.95	0.90
B	0.45	0.65	0.98	0.90	0.75
C	0.95	0.60	0.98	0.30	0.75
D	0.95	0.88	0.98	0.85	0.75

HAPS as Resilient Communication Relays for UAV Operations

Student tasks and acceptance rule

1. Calculate the metrics	Find access-link SINR in dB, end-to-end C2 availability, and mission trust for A–D.
2. Test the backup	Apply the rule below to each case. Explain why the satellite path helps in D and why it would leave the access-link impairment in B unchanged.
3. Rank mission threats	Use the threat inputs on slide 7 to compute each contribution and the total. Identify the largest contributor.

Accept the simulated relay service only if SINR ≥ 6 dB, C2 availability ≥ 0.95, and trust ≥ 0.85.

Submit a metric and decision table for A–D, plus a ranked list of threat contributions.

HAPS as Resilient Communication Relays for UAV Operations

Mission-level relay risk

$$R_{\text{mission}} = \sum_{i=1}^{n} w_i L_i I_i D_i, \qquad (14.46)$$

L = likelihood, I = mission impact, D = detection difficulty. Larger D means harder to detect. Use the simulated 0–1 inputs below.

Threat	Weight w	Likelihood L	Impact I	Difficulty D
UAV access interference	0.40	0.60	0.90	0.70
Ground backhaul outage	0.35	0.50	0.80	0.50
Command replay	0.25	0.30	0.95	0.80

Compute w × L × I × D for each threat, then sum the contributions.

HAPS as Resilient Communication Relays for UAV Operations

Relay performance and decisions

Case	SINR (dB)	C2 availability	Trust	Decision	Failed criteria
A	10.00	0.99	0.9625	Accept	None
B	3.01	0.80	0.7260	Review	SINR, availability, trust
C	10.00	0.78	0.7185	Review	availability, trust
D	10.00	0.95	0.8985	Accept	None

B gives $10 \log_{10}[100 / (5 + 40 + 5)] \approx 3.01$ dB. Changing backhaul leaves this ratio unchanged.

C has a healthy access link but fails service and trust thresholds. In D, satellite backhaul restores C2 availability to 0.95 and mission trust to 0.8985.

HAPS as Resilient Communication Relays for UAV Operations

Worked mission risk and backup checks

Rank	Threat	Weighted contribution
1	UAV access interference	0.1512
2	Ground backhaul outage	0.0700
3	Command replay	0.0570

Total mission risk score = 0.2782. UAV access interference is the largest contributor.

D availability = 1 − (18 + 12 + 0) / 600 = 0.95.

D trust = 0.25 × 0.95 + 0.25 × 0.88 + 0.20 × 0.98 + 0.20 × 0.85 + 0.10 × 0.75 = 0.8985.